



Answers & Solutions:

1. **C. The effective value is the same as that of the original ac wave.**
2. **A. A capacitor in parallel with the rectifier output.**
3. **D. A capacitor in parallel and a choke in series**
4. **C. The output is hard to filter**
5. **C. The filter capacitor(s) must be initially charged**
6. **C. A bleeder resistor**
7. **B. 15.7 V_{rms}**

Solution:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{220 \text{ V}_{\text{rms}}}{V_s} = \frac{14}{1}$$

$$V_s = 15.7 \text{ V}_{\text{rms}} \rightarrow \text{Ans.}$$

8. **D. All of the above are generally needed**
9. **C. 180 degrees**
10. **A. twice that from a half-wave circuit**
11. **B. Parallel with the filter output, reverse-biased**
12. **A. pulsating dc voltage.**
13. **B. 1.45 A, 10.09 %**

Solution:

$$I_L = \frac{V}{R} = \frac{10.9 \text{ V}}{7.5 \Omega}$$

$$I_L = 1.45 \text{ A} \rightarrow \text{Ans.}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

$$VR = \frac{12 - 10.9}{10.9} \times 100\%$$

$$VR = 10.09 \% \rightarrow \text{Ans.}$$

14. **B. A quick-break type**

15. **C. Pulsating dc**

16. **A. 25:1**

Solution:

$$\left(\frac{N_1}{N_2}\right)^2 = \frac{R_L}{R_L'} = \frac{10 \text{ k}\Omega}{16 \Omega} = 625$$

$$\frac{N_1}{N_2} = \sqrt{625} = 25:1 \rightarrow \text{Ans.}$$

17. **D. A capacitor in parallel with the dc output**

18. **B. A TV broadcast transmitter**

19. **D. Use two capacitor/choke sections one after the other**

20. **A. It uses the whole transformer secondary for the entire ac input cycle**

21. **A. The primary voltage is greater than the secondary voltage**

22. **C. Has one tapped winding**

23. **B. Half of the ac wave is chopped off.**

24. **A. Provides maximum coupling**

25. **D. The balanced winding**

26. **C. capacitor**

27. **A. Diode failure**

28. **C. low-pass filter**

29. **B. Full wave, center-tap**

30. **B. The I²R losses are lower**

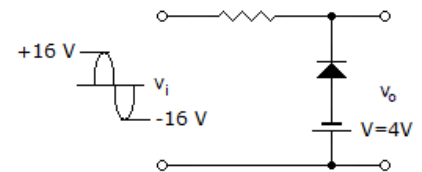
31. **C. 40:1**

Solution:

$$\frac{N_p}{N_s} = \sqrt{\frac{Z_p}{Z_s}} = \sqrt{\frac{12.8 \text{ k}\Omega}{8 \Omega}} = \frac{40}{1} \rightarrow \text{Ans.}$$

32. **C. The filter**

33. **A. 16 V, -4 V**



Solution:

During the positive half-cycle:
Diode-Reverse Bias

$$I = 0$$

$$V_o = 16 \text{ V}$$

During the negative half-cycle:
Diode-Forward Bias

$$V_o = 0 - 4 \text{ V}$$

$$V_o = -4 \text{ V}$$

34. **B. There will be an impedance mismatch, no matter what the turns ratio of the transformer**

35. **B. rectifier**

36. **C. The fuse will blow out right away**

37. **B. A step-down unit**

38. **C. Low-pass filter**

39. **C. The primary voltage is less than the secondary voltage**

40. **D. Poor regulation**

41. **B. The effective value is less than that of the original ac wave.**

42. **D. 53.6 V**

Solution:

$$V'_{dc} = \frac{R_L}{R + R_L} V_{dc}$$

$$V'_{dc} = \frac{1000 \Omega}{120 \Omega + 1000 \Omega} (60 \text{ V})$$

$$V'_{dc} = 53.6 \text{ V} \rightarrow \text{Ans.}$$



43. C. An increase in core loss

44. A. Connected in parallel with filter capacitors

45. C. 2

46. A. Placing the windings on opposite sides of a toroidal core.

47. B. 16.96 V

Solution:

$$\begin{aligned}V_{dc} &= V_p - \frac{I_{dc}}{4fC} \\&= 18V - \frac{100mA}{4(60Hz)(400\mu F)} \\&= 16.96V \rightarrow \text{Ans.}\end{aligned}$$

48. C. ripple

49. A. 63.6 V

Solution:

$$\begin{aligned}V_{dc} &= 0.318V_m \\V_{dc} &= 0.318(200) \\V_{dc} &= 63.6V \rightarrow \text{Ans.}\end{aligned}$$

50. C. Leave it alone and have a professional work on it

END